Huawei Access Node JAY-ACC-39 Hardware/Environmental Fault Resolution SOP

Node: JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1926

Description

This Standard Operating Procedure outlines the steps to identify and resolve Hardware/Environmental Faults on the Huawei Access Node JAY-ACC-39 at the Jayanagar Metro Site. The failure domain of interest is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, manifesting as ALARM_1926 with a severity of Warning. In the Huawei Access Node, the Line Card (LC) is responsible for terminating the optical fibers and converting the optical signals to electrical signals. The LC's reliability is critical to the overall network's availability. However, due to the nature of the components and the environment in which they operate, Hardware/Environmental Faults can occur. These faults can arise from various sources, including but not limited to, manufacturing defects, exposure to extreme temperatures, humidity, or power surges. The symptoms of these faults may manifest as errors in the ASIC memory, line card crashes, or loss of power supply redundancy. To mitigate these issues, the Huawei Access Node is designed with redundant components and mechanisms to detect and isolate faults. The Line Card, for instance, has a built-in watchdog timer that monitors its own health and triggers a self-test upon detecting anomalies. Additionally, the Node's management system continuously monitors the health of the Line Card and other components, triggering alarms when anomalies are detected. This SOP outlines the steps to identify the root cause of the Hardware/Environmental Fault, isolate the affected component, and implement corrective actions to restore the Node's functionality.

Pre-requisite Checks

- Verify that the Node's power supply is functioning correctly and that there are no signs of physical damage or tampering.

- Confirm that the Node's redundant components are online and functioning as expected.

- Ensure that the Node's software is up-to-date and that there are no pending software updates.

- Dispatch a remote hands team to the Jayanagar Metro Site to physically inspect the Node and its components.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration, Expected Output: Display the current configuration of the Node, including the Line Card's configuration., Description: This command will help identify any configuration-related issues that may be contributing to the Hardware/Environmental Fault.

Command: show alarm all, Expected Output: Display all alarms on the Node, including the ALARM_1926., Description: This command will help identify the root cause of the Hardware/Environmental Fault and any associated alarms.

Command: display hardware, Expected Output: Display the Node's hardware configuration, including the Line Card's hardware configuration., Description: This command will help identify any hardware-related issues that may be contributing to the Hardware/Environmental Fault.
```

Resolution Steps

Step 1: Step: Isolate the affected Line Card, Command: undo linecard slot , Description: This command will isolate the affected Line Card and prevent it from further affecting the Node's functionality.

Step 2: Step: Swap the Line Card, Command: switchover linecard slot , Description: This command will swap the affected Line Card with a spare Line Card, restoring the Node's functionality.

Step 3: Step: Verify the Node's functionality, Command: display current-configuration, Description: This command will help verify that the Node's functionality has been restored and that the Hardware/Environmental Fault has been resolved.

Post-Check Verification

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Command: display current-configuration, Expected Output: Display the current configuration of the Node, including the Line Card's configuration., Description: This command will help verify that the Node's configuration has been updated correctly.

Command: show alarm all, Expected Output: No ALARM_1926., Description: This command will help verify that the Hardware/Environmental Fault has been resolved.
```

Roll-back Procedures

- Step: Revert the Line Card swap, Command: undo switchover linecard slot , Description: This command will revert the Line Card swap and restore the original Line Card.

- Step: Re-enable the affected Line Card, Command: enable linecard slot , Description: This command will re-enable the affected Line Card and restore its functionality.